Chapter 13, Problem 66P

Problem

Design a problem to help other students better understand how the ideal autotransformer works.

Step-by-step solution

Step 1 of 1

Design a problem to help other students better understand how the ideal autotransformer works.

Although there are many ways to solve this problem, this is an example based on the same kind of problem asked in the third edition.

Problem

An ideal autotransformer with a 1:4 step-up turns ratio has its secondary connected to a 120-load and the primary to a 420-V source. Determine the primary current.

Solution

$$v_1 = 420 \text{ V} \qquad (1)$$

$$v_2 = 120 I_2 \qquad (2)$$

$$v_1/v_2 = 1/4 \text{ or } v_2 = 4v_1 \qquad (3)$$

$$I_1/I_2 = 4 \text{ or } I_1 = 4 I_2 \qquad (4)$$

Combining (2) and (4),

$v_2 = 120[(1/4)I_1] = 30 I_1$

$4v_1 = 30I_1$

$4(420) = 1680 = 30I_1$ or $I_1 =$ **56 A**